

This analysis uses weekly stock data for **Tesla Inc.** and **Nike Inc.** to evaluate their investment characteristics and the relationship between the two assets. The dataset includes weekly returns, summary statistics, covariance and correlation analysis, regression models, and statistical tests. These methods help observe how the returns of each stock change over time and assess their risk, variability, and relationship with the overall market.

The summary statistics provide key information about the behavior of both stocks. Tesla has an average weekly return of approximately **0.0103**, indicating a positive average performance during the analyzed period, while Nike has a slightly negative mean return of about **-0.0042**. The standard deviation of returns is **0.0624 for Tesla** and **0.0554 for Nike**, suggesting that Tesla's returns are slightly more volatile. This indicates that Tesla carries a higher level of risk compared to Nike, since its returns fluctuate more widely from week to week.

The relationship between the two assets was evaluated using covariance and correlation analysis. The results show a **very weak positive relationship** between Tesla and Nike returns. This means that while the stocks sometimes move in the same direction, the connection between them is not strong. Because of this weak relationship, combining the two stocks in a portfolio may provide **diversification benefits**, helping reduce overall investment risk.

Regression analysis was conducted to examine how each company's stock return responds to changes in the overall market return. In this model, market return is used as the independent variable while the individual stock return is the dependent variable. The regression results suggest that market movements influence the returns of both stocks, although the strength of this relationship varies. Tesla tends to show higher sensitivity to market fluctuations, meaning its returns may react more strongly to market changes compared to Nike.

Finally, statistical tests were performed to evaluate the variability of the stock returns. A **chi-square test** was used to determine whether the population variance differs from a hypothesized annual standard deviation of **25%**, based on the sample variance of weekly returns. An **F-test** was also conducted to compare the variances of Tesla and Nike. The results show that **Tesla has a higher sample variance (0.00389) than Nike (0.00306)**, indicating that Tesla's returns exhibit greater variability. This confirms the earlier findings from the summary statistics that Tesla is the more volatile stock of the two.